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Using Data Mining Techniques for Detecting Terror-Related Activities on the Web

Abstract:

The number of terror attacks is increasing year by year. On November 13, 2015, the terrorist attack that took place in Paris caused hundreds of deaths. The hazards of cyber terrorism have already become more and more serious. The USA has enacted a number of laws aimed at the prevention of cyber terrorism, such as “USA PATRIOT Act”. It is necessary to establish a model for the prevention of terrorist network spread and to monitor and find the people with a tendency to terrorism. The Internet behavior analysis and risk assessment model (IBARA) was established for the Internet to assess the internet behaviors of those people who are monitored. In this paper, based on IBARA, we not only research the relationship between people’s Internet behavior and their possible terrorist tendency, but also analyze and discuss the relative quantitative risk index of individual terrorism tendency and the relevant strategies to prevent terrorist attacks.

Firstly, the Internet behavior was divided into two parts: Web text and image. The complex vector space of word frequency analysis algorithm was adopted to establish the personal tendency of terrorism risk index sub module (PTTRISM) which can predict people’s tendency to terrorism. In PTTRISM, this paper analyzes the behavior of individual Web text using the keyword extraction technique and frequency analysis technique. According to the analysis results, it’s given the value of the risk index of individual terrorism in this paper. Using the PTTRISM to analyze the data sample, we had drawn a conclusion that most people who have been access to the terrorism-related information are not likely to become potential terrorists. The PTTRISM could calculate people’s risk index about the tendency to terrorism through analyzing Internet behavior.

Secondly, in fact, the object of network monitoring is not a person but a large number of people, which makes to monitoring data too large and complex. In order to facilitate the rapid and efficient classification and analysis of big data, a big data clustering statistics sub module (MDCSSM) is established based on the technique of density-based clustering. At the same time, in order to shorten the computing time of the MDCSSM, in this paper is adopted the standard particle swarm optimization (PSO) with the weight-shrink factor. It realized the effective, fast and automatic clustering analysis of datasets. Validation of the sub model using the data, The model can be used to analyze a large amount of data. Due to sacristy of the monitoring data, we utilize some frequently-tested public datasets, “Iris”, “Glass”, “Wine” and “Aggregation” to replace the monitoring data and verify the clustering algorithm. The clustering results demonstrate that the clustering algorithm can categorize the monitoring datasets in an effective, fast and automatic manner.

Finally, We propose some suggestions to President Obama about fighting against terrorism as follows based on IBARA :

- 1.Put into more resources in terms of network against terrorism. You could build User Online Monitoring System of Behavior and Psychological to monitor and assess the behavior of the public.
- 2.Establish Information security evaluation system to weaken and even prevent the terrorist propaganda through the network.
- 3.Strengthen public anti-terrorism education, raise public awareness of anti-terrorism.

Due to the time constraints, the model still has some defects which need to be improved. In the PTTRI sub module, factors of voice and image files are not considered. In the MDACS sub module, the selection of adaptive function in Clustering analysis could be further improved. With the further improvement of the model, we will get more accurate results.

Key words: PSO, word frequency analysis algorithm , density-based clustering, terrorism, Internet behavior

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I. Introduction

In order to indicate the origin of web-related terrorism problems, the following background is worth mentioning.

Terrorist cells are using the Internet infrastructure to exchange information and recruit new members and supporters^{[1][2]} (Lemos 2002; Kelley 2002). For example, high-speed Internet connections were used intensively by members of the infamous ‘Hamburg Cell’ that was largely responsible for the preparation of the September 11 attacks against the United States^[3] (Corbin 2002). This is one reason for the major effort made by law enforcement agencies around the world in gathering information from the Web about terror-related activities. It is believed that the detection of terrorists on the Web might prevent further terrorist attacks^[2] (Kelley 2002). One way to detect terrorist activity on the Web is to eavesdrop on all traffic of Web sites associated with terrorist organizations in order to detect the accessing users based on their IP address. Unfortunately it is difficult to monitor terrorist sites^[3] (such as ‘Azzam Publications’ (Corbin 2002) since they do not use fixed IP addresses and URLs. The geographical locations of Web servers hosting those sites also change frequently in order to prevent successful eavesdropping. To overcome this problem, law enforcement agencies are trying to detect terrorists by monitoring all ISPs traffic^[4] (Ingram 2001), though privacy issues raised still prevent relevant laws from being enforced.

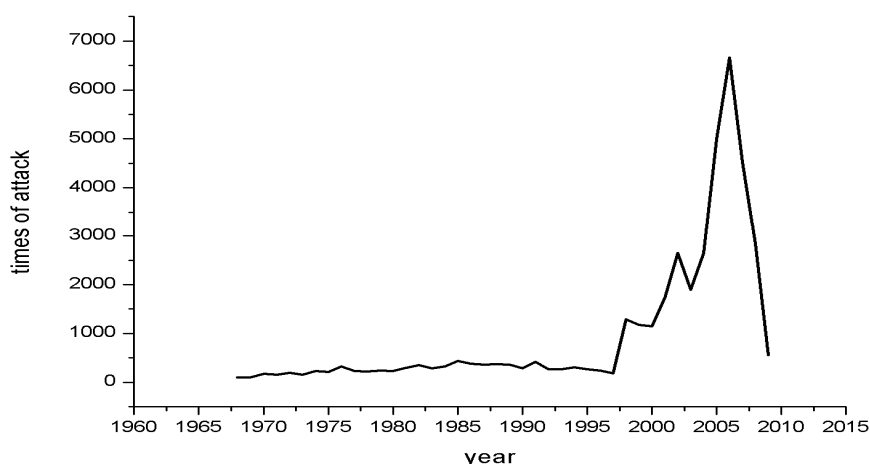


Figure 1: the annual number of terrorists' attack from 1968 to 2009

II. The Description of the Problem

2.1 Our Approximation the Whole Course of Data Mining To terrorists on websites

- I How often does the internet user who is monitored visit the website that contains terrorized information and propaganda of terrorism.
- I The lexical meaning of contents of their emails, chats, post views and text files being downloaded.

- I As for other formats of files, such as videos, images and audios, the techniques of the image description and voice recognition are used as a tool to detect the terrorists.
- I For categorizing the monitoring data, the cluster techniques are adopted to select data in an effective, fast and automatic manner.
- I Present some useful suggestions to President Obama for fighting terrorism

2.2 The Differences in Weights and Sizes of Available Data

Due to differences between the collected datasets, it's quite necessary to preprocess the available data, Such as text datasets, numerical datasets, image datasets and even voice datasets.

- 1) The Preprocess of Text Data: remove non-alphabetical characters from the text dataset and put them into MATLAB cell structures.
- 2) The Preprocess of Image Data: remove non-imagery information from the image datasets and convert the RGB images into the gray-value images. If the image datasets are polluted by noises, it's quite necessary to denoise image before analyzing the relevant information.
- 3) The Preprocess of Voice: if the audio datasets are polluted by noises, it's a need to implement audio-denoising steps before digging out the auditory information.
- 4) The Preprocess of Numerical Dataset: Due to existence of differences between data samples in units and magnitudes, the numerical dataset needs to be normalized and standardized.

III. IBARA

3.1 PTTRISM

In this paper a new methodology to detect users accessing terrorist related information by Frequency-Analysis Techniques, Vector Space Models of Lexical Meaning^[5], Image Description^[6] and Voice Recognition^[7], Data Cluster^[8].

3.1.1 Terms, Definitions and Symbols in PTTRISM

The signs and definitions are mostly generated from our models in this paper.

- I R is the risk index, which denotes the risk degree that the Internet user can be.
- I P_t^{text} is the degree that the text contents that the Internet user involves are related to terrorism during the time interval t .
- I P_t^{image} is the degree that images that the Internet user browses and download are related to terrorism during the time interval t .
- I P_t^{audio} is the degree that audios that the Internet user listens to and download are related to terrorism during the time interval t .
- I $w_{i,j}$ is the weight factor of vector space $\vec{q}$.

3.1.2 Assumptions in PTTRISM

The main design criteria for the proposed methodology are:

- ┆ Training the detection algorithm should be based on the content of existing terrorist sites and known terrorist traffic on the Web.
- ┆ Detection should be carried out in real-time. This goal can be achieved only if terrorist information interests are presented in a compact manner for efficient processing.
- ┆ The detection sensitivity should be controlled by user-defined parameters to enable calibration of the desired detection performance.
- ┆ All information related to terrorism is not encrypted by enciphered algorithms, such as RSA
- ┆ All information that can be monitored is presented by images, audios and texts.
- ┆ Neglect the social attributes of the monitored person and only consider the network properties

3.1.3 The Model of Terrorism-Related Website Browsing and Vector Space Models of Lexical Meaning

One major issue in this model is the representation of textual content of Web pages. More specifically, there is a need to represent the content of terror-related pages as against the content of a currently accessed page in order to efficiently compute the similarity between them^[9]. This study will use the vector-space model commonly used in Information Retrieval applications for representing terrorists' interests and each accessed Web page. In the vector-space model, the weight $w_{i,j}$ associated with a pair (k_i, d_j) is positive and non-binary. Further, the index terms in the query q are also weighted. Let $w_{i,q}$ be the weight associated with the pair (k_i, q) where $w_{i,q} \geq 0$. Then, the query vector $\vec{q} = (w_{1,q}, w_{2,q}, \mathbf{K}, w_{t,q})$ is defined as where t is the total number of index. The vector for a document d_j is represented by $\vec{d}_j = (w_{1,j}, w_{2,j}, \mathbf{K}, w_{t,j})$. The vector model proposes to evaluate the degree of similarity of the document d_j with regard to the query q as the correlation between the vectors $\vec{d}_j$ and $\vec{q}$. This correlation can be measured by the cosine of the angle between these two vectors as,

$$sim = \frac{\vec{d}_j \cdot \vec{q}}{|\vec{d}_j| |\vec{q}|} \quad (3-1-1)$$

Where $|\vec{d}_j|$ and $|\vec{q}|$ are the norms of the document and query vectors. In the vector space model, the frequency of a term k_i inside a document d_j

$$freq_{i,j} = \frac{n_i}{N_j} \quad (3-1-2)$$

The normalized frequency of term k_i inside a document d_j is given by

$$f_{i,j} = \frac{freq_{i,j}}{\max(freq_{i,j})} \quad (3-1-3)$$

The best known term-weighting schemes use weights which are given by

$$w_{i,j} = -f_{i,j} \log(freq_{i,j}) \quad (3-1-4)$$

In this paper each Web page is considered as a document and is represented as a vector. The terrorists' interests are represented by several vectors where each vector relates to a different topic of interest. The query of the methodology defines and represents the typical behavior of terrorist users based on the content of their Web activities. The query is based on a set of Web pages that were downloaded from terrorist related sites and is the main input of the detection algorithm. It is assumed that it is possible to collect Web pages from terror-related sites. The content of the collected pages is the input to the Vector Generator module that converts the pages into vectors of weighted terms^[10] (each page is converted to one vector).

In order to define the degree that the internet user browses the terrorism-related websites during the time interval t , the formula $b(m)$ is defined by the function that the internet user behaves like a potential terrorist when browsing the website m as follows:

$$b(m) = \text{simC}(\text{sim}) \quad (3-1-5)$$

$$\text{where } c(x) = \begin{cases} 1 & x \geq \text{threshold} \\ 0 & x < \text{threshold} \end{cases}$$

$$P_t^{\text{text}} = \frac{\sum b(m)}{M} \quad (3-1-6)$$

In this paper, we adopt 0.5 as the value of threshold. The query in this paper is listed in the table below.

ID	Details of Queries
1	Bomb Suicide
2	Gunfire
3	Kidnap
4	Massacre
5	Attack to Civilians
6	Islamic State of Iraq and al Shams
7	Qaeda
8	al-Shabaab
9	Islamic State
10	hijack
11	Assassination

3.1.4 The Model of Risk Index

Here we report the remarkable finding that identical patterns of violence are currently emerging within these different international arenas. Not only have the wars in Iraq and Colombia evolved to yield a same power-law behavior, but this behavior is

currently of the same quantitative form as the war in Afghanistan and global terrorism in non-G7 countries. Not only is the model's power-law behavior in excellent agreement with the data from Iraq, Colombia and non-G7 terrorism, it is also consistent with data obtained from the recent war in Afghanistan. Power-law distributions are known to arise in a large number of physical, biological, economic and social systems. In the present context, a power-law distribution means that the probability that an event will occur with behavior P is given by^[12]

$$R(P) = CP^{-\alpha} \quad (3-1-7)$$

where $P \in (0,1]$, $P = P_t^{text}$ and C and α are positive coefficients^{[13][14]}, Previous studies have shown that the distribution obtained from past terrorists' attack exhibits a power-law with^[15] $\alpha = 1.809$.

Since we can't get the coefficients C effectively, we define a relative risk index r among a group of people who are monitored during the specific time interval as follows

$$r_a = \frac{R_a(P)}{\bar{R}_a(P)} \quad (3-1-8)$$

3.1.5 Solutions and Results for PTTRISM

1) The Solution Steps to PTTRISM

1. Generation of Term-Frequency matrix

It is term-frequency matrix of all unique terms in document d_j with $j = 1, 2, \dots, K, N$.

The term document matrix $Freq$ is a $M' \times N$ matrix with t_i unique terms in dictionary $i = 1, 2, \dots, K, M$ and N documents the elements of $Freq$ are represented as in $freq_{i,j}$ which each element indicates the frequency of i^{th} term in j^{th} document.

The Cranfield data collection is preprocessed to convert into individual 1398 text files. Also, non-embedding special characters and numerals have been removed from these files. 79,728 words have been collected which are then processed to find the frequency of unique words in each documents. The dictionary of unique words is of 7805 words. Thus the term frequency matrix is of size $7805' \times 1398$.

2. Generation of Query matrix and Term-weight calculations and result

After removing stop-list words and non-embedding special characters is used as query, which contributes to the set of 1398 unique queries represented as q . Here, we have taken queries as titles of the document instead of the dataset queries so as to judge the relevancy more profoundly. The generated matrix for 1398 queries is $Q_{1398' \times 7805}$. A term-frequency matrix is processed to get the term weights considering term-weighting schemes.

2) The Results of PTTRISM

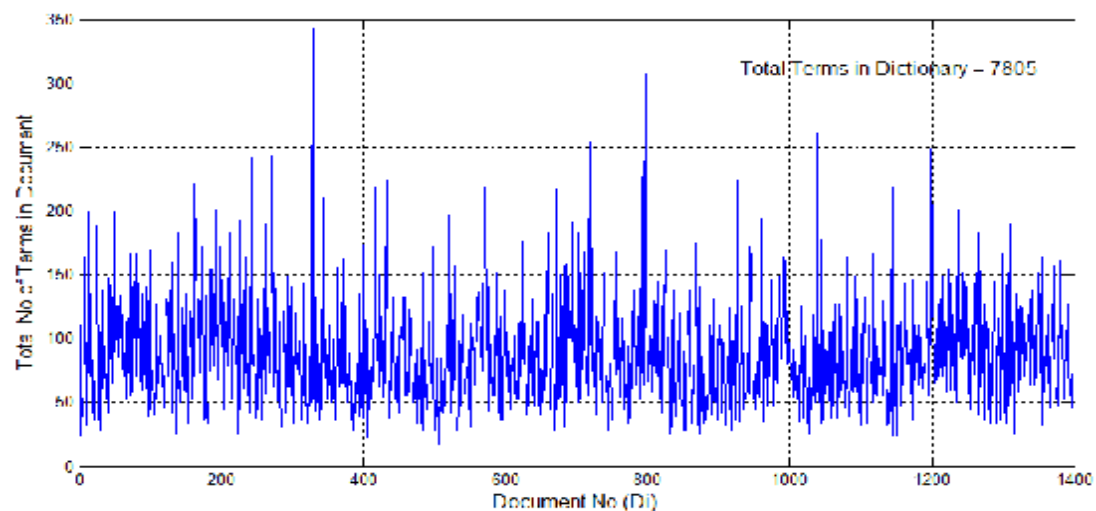


Figure 2: Index terms in a dictionary

Figure 2 shows the distribution of index terms in dictionary for individual documents. The dictionary consists of 7805 unique terms.

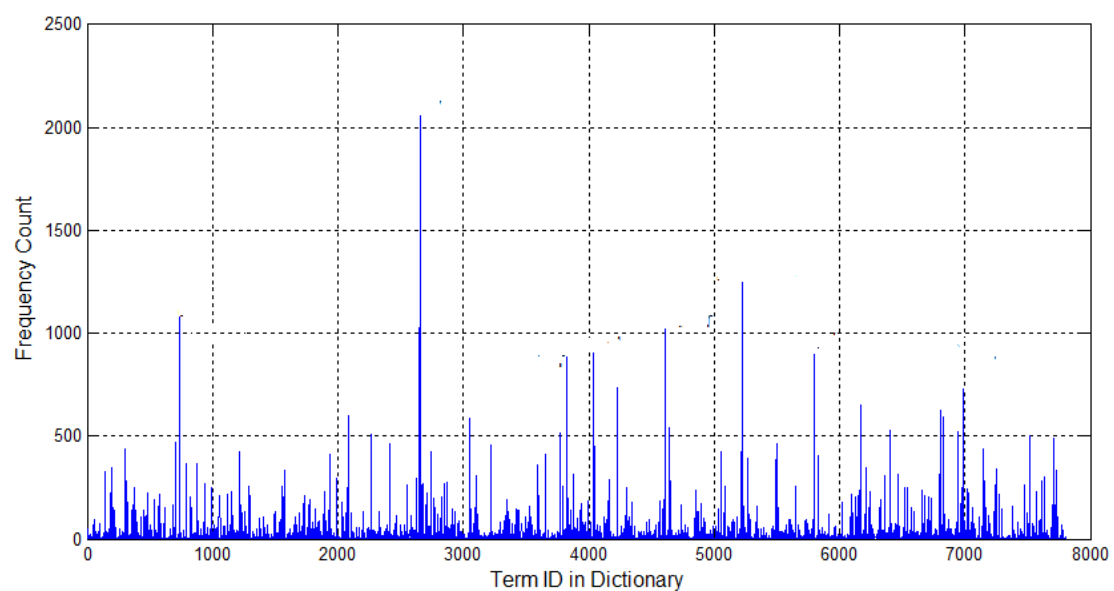


Figure 3: Frequency count of each unique term among data collection

Figure 3 shows frequency count of each unique term in dictionary distributed in complete dataset. Some of the unique terms such as ('ISIS', 2059), ('Qaeda', 1245), ('hijack', 1076), ('Assassination', 897), with high frequency in entire documents is shown.

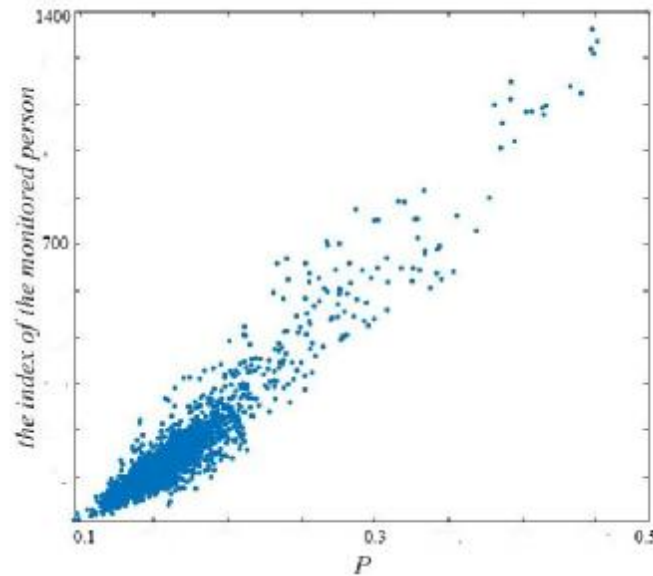


Figure 4: the distribution of the P value among the monitored persons

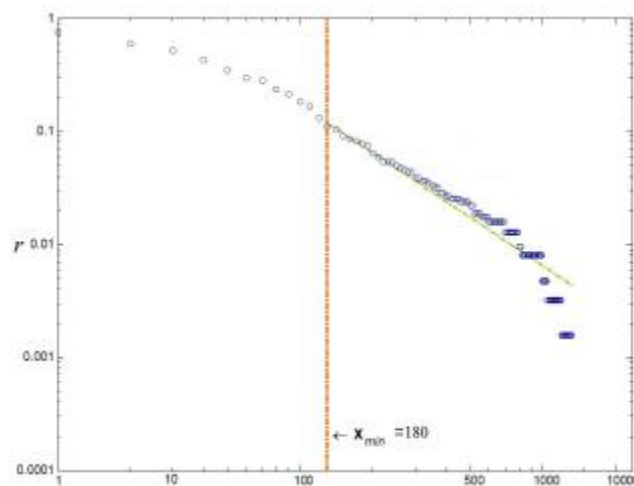


Figure 5: the distribution of the r value among the monitored persons

In the Figure 5, we define 0.1 as the threshold of the risk index. If a one's risk index is beyond 0.1, he or she can become a potential terrorist, and otherwise more likely to be an ordinary person

From the 1398 individual text files that are obtained from 1398 individuals, we can easily draw a conclusion that most people who have been access to the terrorism-related information are not likely to become potential terrorists. There are just 12 ones of all monitored persons who are likely to become potential terrorists, besides all their risk indexes are beyond 0.1.

3.1.6 Strength and Weakness in PTTRISM

- I **Strength:** The model has proved that we can detect and discriminate terrorists from many internet users who are monitored for the purpose of preventing terrorism. Moreover, we have drawn some useful conclusions about the tendency of terrorism. Through the calculation of the model, we can find that most people who are monitored and have access to the

information of terrorism will not be likely become a potential terrorist. The model can be applied to such those activities, terrorist surveillance and criminal activity monitoring.

- I **Weakness:** This model just applies to analyze the text information of those people who were monitored. As we have stated in the article, the network behavior of the monitored person may be analyzed from three aspects of information, that is, text, image and website. In our model, we use only the frequency analysis of the text to evaluate the risk index of the person who is monitored on real-time.

3.2 MDCSSM

Experts use the expressive big data to indicate huge amounts of information. Especially in the contemporary informational world, big data can reveal a lot of information that is hidden behind the big data. The analytical techniques of big data can be utilized to fight against terrorism. But unlike the traditional data mining, data analysis of terrorism-related information needs categorizing them in an effective, fast and automatic manner. In this paper, we present a new clustering algorithm to categorize big data.

3.2.1 Extra Symbols

Signs and definitions indicated above are still valid. Here are some extra signs and definitions.

- I v_i^t is the velocity of particle i at iteration t .
- I x_i^t is the position of particle i at iteration t .
- I $rd_{i,j}$ is the distance between the sample i and the sample j
- I $(\hat{x}, \hat{y})$ is the relative coordinate pair based the $rd_{i,j}$

3.2.2 Additional Assumptions

- I The test dataset in the paper can be equivalent to the monitoring dataset.
- I The clustering algorithm can obtain how many groups of the dataset and how many remainders in each group, but can't achieve to obtain the information of each group.

3.2.3 The Foundation of MDCSSM to Categorize Big Data

In terms of clustering's objectives, the aims of data clustering are minimize similarity among instance in one cluster and maximize dissimilarity between two clusters.

Particle Swarm Optimization (PSO) is a population-based search algorithm which simulates bird behavior in finding food^[16]. In PSO, birds represented as particle are employed to perform solution exploration. Particle at any initial position moves to other position with a certain velocity and direction. In every iteration velocity is adjusted based on particle best solution and social best solution. The velocity of each particle is calculated according to formula shown in Equation (3-2-1), while particle's position is updated using Equation (3-2-2). This concept makes PSO becomes different with other heuristic methods^[17].

$$v_i^{t+1} = wv_i^t + c_1r_1(pb_{best_i} - x_i^t) + c_2r_2(g_{best} - x_i^t) \quad (3-2-3)$$

$$x_i^{t+1} = v_i^{t+1} + x_i^t \quad (3-2-4)$$

Where v_i^t and x_i^t are velocity and position of particle i at iteration t , respectively. The notation w is defined as inertia weight. This is a mechanism to control the exploration and exploitation abilities of the particle and also to eliminate the need for velocity clamping. In this paper, PSO algorithm is extended so that it can overcome automatic clustering problem. Generally, there are two major issues in solving automatic and fast clustering, namely determining cluster number and finding cluster centroids. In order to accommodate these two objectives, the automatic clustering algorithm employs a particle which comprises of two sections. The first section is responsible for representing number of clusters while the other section informs cluster centroids. Exploration step is conducted towards these two sections gradually. After updating a section, infeasibility constraint will be checked^[18]. It covers feasible number of clusters and cluster's size. In addition, k -means is performed to update the second section of each particle based on active clusters obtained by its first section.

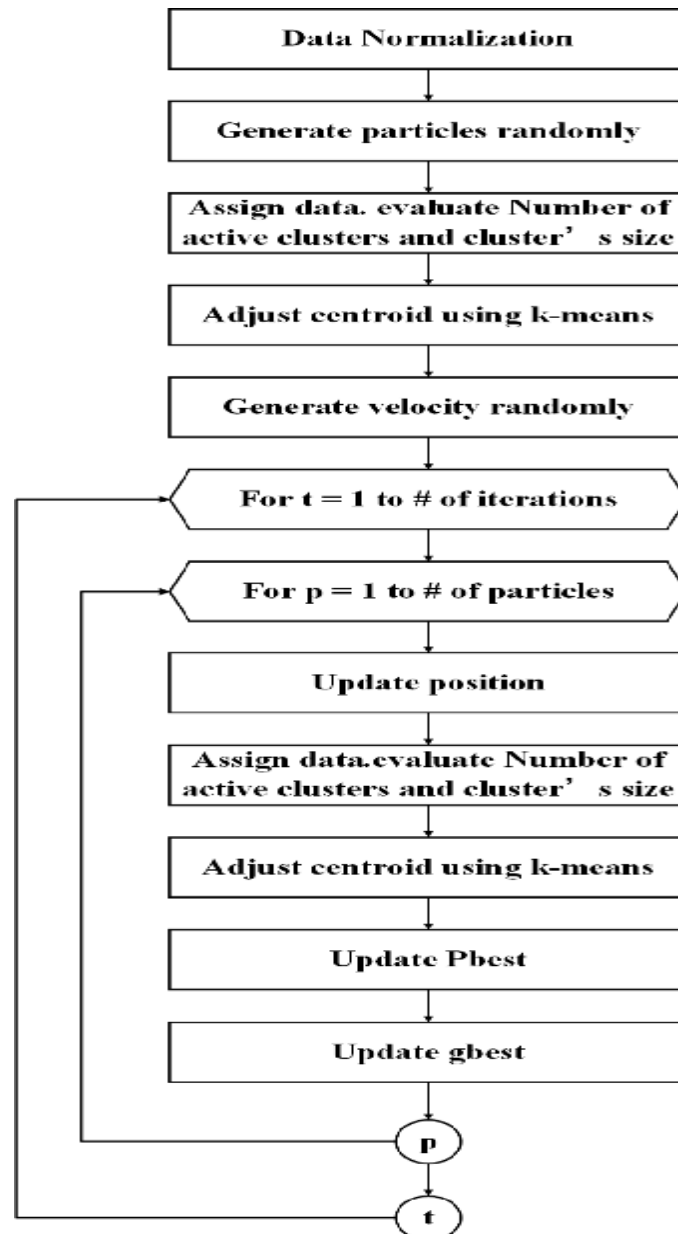


Figure 6: flow chart of the model 2

A particle comprises of two sections. Section one represents number of clusters while section two gives information regarding cluster centroid or data points belongs to the corresponding cluster. Through this scheme, each particle has its own exploration path so that the search space can be explored wider and deeper. Supposes data clustering is conducted towards a given dataset $D = \{d_j | j = 1, 2, \dots, K, S\}$ where S is number of data points and it consists of M attributes. Let the appropriate number of clusters is an integer number in the range $[2, N_{\max}]$, where $N_{\max}$ is predicted maximum number of clusters. A particle p in iteration t symbolized as X_p^t comprises of $(1+M)N_{\max}$ bits as illustrated as the following chart^[19].

X_{p1}^t	L	$X_{pN_{\max}}^t$	$X_{pN_{\max}+1}^t$	L	$X_{pN_{\max}+M.N_{\max}}^t$
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All bits are continuous number within $[0,1]$. The first $N_{\max}$ bits represents active or inactive clusters. Cluster k in particle p is an active cluster if the k^{th} bit in X_p^t is greater than or equal to 0.5. Otherwise, the corresponding cluster is inactive. Furthermore, the rest bits correspond to cluster centroid. The initial particle and velocity are generated randomly. Each bit i in a particle is updated using Equation (3-2-5) and Equation (3-2-6). Furthermore, an additional procedure is embedded to guarantee feasibility of the next particle. In this additional procedure,

$$v_{pi}^{t+1} = w^t v_{pi}^t + c_1 r_1 (y_{pi} - x_{pi}^t) + c_2 r_2 (z_i - x_{pi}^t) \quad (3-2-7)$$

$$x_{pi}^{t+1} = x_{pi}^t + v_{pi}^{t+1} \quad (3-2-8)$$

In this additional procedure, if result from Equation (3-2-9), is outside $[0,1]$, sigmoid function is employed to transform the current result to the feasible region.

$$x_{pi}^{t+1} = \begin{cases} \text{sigmoid}(x_{pi}^{t+1}), & \text{if } x_{pi}^{t+1} \notin [0,1] \\ x_{pi}^{t+1}, & \text{otherwise} \end{cases} \quad (3-2-10)$$

there:

v_{pi}^t : velocity of the i^{th} bit in particle p for iteration t .

y_{pt} : the i^{th} bit in particle best p .

z_i : the i^{th} bit in global best.

In updating the velocity, two parameters are required. There are inertia weight w , and learning rate c_1 and c_2 . The clustering algorithm applies time-varying inertia weight. Herein, the inertia weight linearly decreases based on the Equation (3-2-11),

$$w = w_{\max} - \frac{(w_{\max} - w_{\min})}{T} t \quad (3-2-12)$$

here the T is total iteration and t is current iteration.

1) Procedure for evaluating number of clusters^[20].

Given particle $X_p^t = \{x_{pi}^t, \dots, x_{pN_{\max}+MN_{\max}}^t\}$, N_p are active clusters in X_p^t , and $|N_p|$ is number of active clusters.

If $|N_p| < 2$

Generate the integer number $i, j \in \{1, \dots, N_{\max}\}, i \neq j$

If $x_{pi}^t < 0.5$

$$x_{pi}^t = x_{pi}^t + 0.5$$

$$|N_p| = |N_p| + 1$$

End if

If $x_{pj}^t < 0.5$

$$x_{pj}^t = x_{pj}^t + 0.5$$

$$|N_p| = |N_p| + 1$$

End if

End if

2) Procedure for evaluating cluster's size^[20]

Given particle $X_p^t = \{x_{pi}^t, \dots, x_{pN_{\max} + MN_{\max}}^t\}$, N_p is active clusters in X_p^t , and $|N_p|$ is the number of active clusters, and $|x_{pi}^t|$ is size of cluster x_{pi}^t

For $i=1$ to $N_{\max}$

If $x_{pi}^t \geq 0.5$

If $|x_{pi}^t| < 2$ and $|N_p| > 2$

$$x_{pi}^t = x_{pi}^t - 0.5$$

If $|x_{pi}^t| = 1$

Reassign data to another cluster which has the shortest distance

End if

$$|N_p| = |N_p| - 1$$

Else if

Choose any data points $d_q, d_r \in D, q \neq r$

Let x_{pi}^t be the original cluster of d_q and d_r

Assign data d_q, d_r to cluster x_{pi}^t

$$\text{Update } |x_{pi}^t| = |x_{pi}^t| + 2$$

$$\text{Update } |x_{pj}^t| = |x_{pj}^t| - 2$$

End if

End if

End for

In this paper, the fitness function of PSO clustering algorithm is given as follows

$$f_{\text{fitness}} = (0.5 \frac{1}{\sqrt{2ps^2}} e^{-\frac{(N_p - m)^2}{2s^2}} + 1) \frac{\min_{i,j \in N_p, i \neq j} \{ \|c_i - c_j\|^2 \}}{\text{iteration}} \quad (3-2-13)$$

3.2.4 The Results of MDCSSM

Since the scarcity of the big monitoring data, other datasets are adopted to test our clustering algorithm of particle swarm optimization, such as “Iris”, “wine”, “Glass” and “Aggregation” from the public dataset websites. After finishing clustering to data, the categorized data needs to be visualized in a direct manner. For one-dimensional or

two-dimensional, even three dimensional data, the dataset can be directly visualized in the Cartesian coordinate. But for high dimensional, the dataset can't be visualized in the Cartesian coordinate, thus a technique of dimension-reduction is introduced in this paper, the multidimensional-scaling method is introduced as follows^[21]:

$$rd_{i,j} = \sqrt{\sum_{k=1}^M (d_{i,k} - d_{j,k})^2} \quad (M > 3) \quad (3-2-14)$$

$$[x, y] = \underset{i, j}{\operatorname{argmin}} \sum_i^g \sum_j^g ((x_i - x_j)^2 + (y_i - y_j)^2 - rd_{i,j}^2) \quad (3-2-15)$$

$s.t. \ x_i \geq 0, x_j \geq 0$

In the dataset Iris, there are 150 samples with 4 attributes. Thus, it's a need to demonstrate the cluster centers and cluster results in two different graphs.

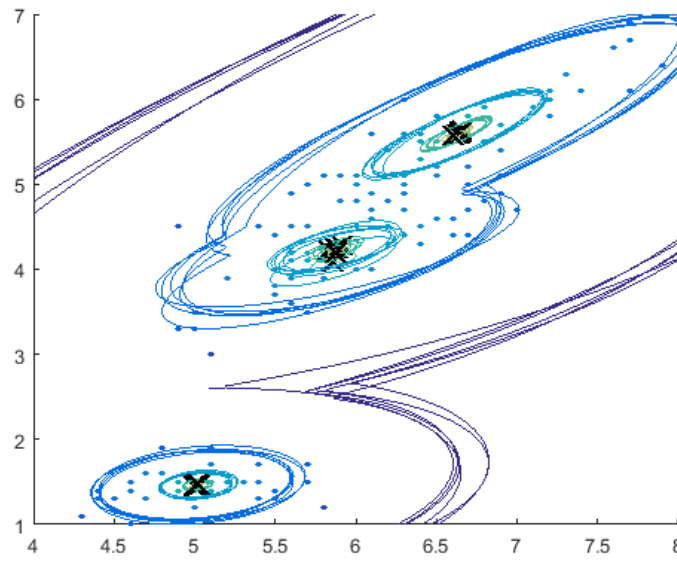


Figure 7: Cluster centers Iris dataset in a relative coordinate system

From the Figure 7 it can be easily concluded that there are three clustering centers in the dataset Iris, which means the dataset Iris can be grouped into three categories.

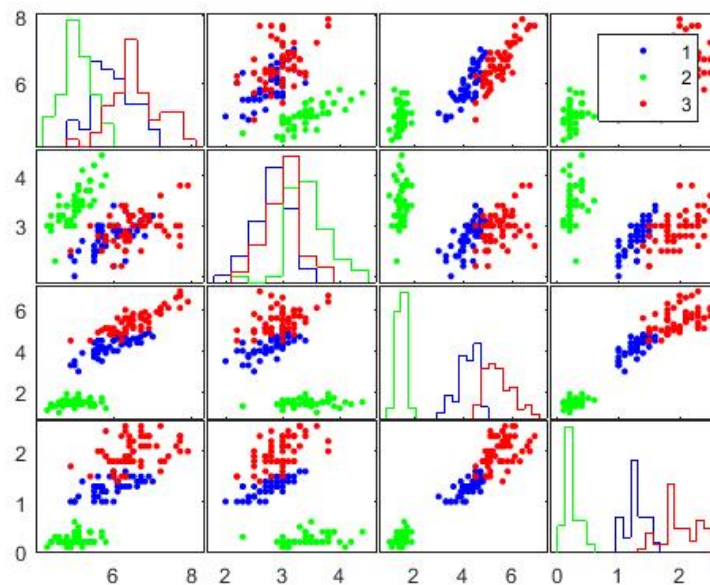


Figure 8: Cluster distribution of Iris data utilizing PSO clustering algorithm

From the Figure 8, the dataset Iris can be separated into three groups as the particle swarm optimization goes on like the four step-shaped sub graphs in the Figure 7.

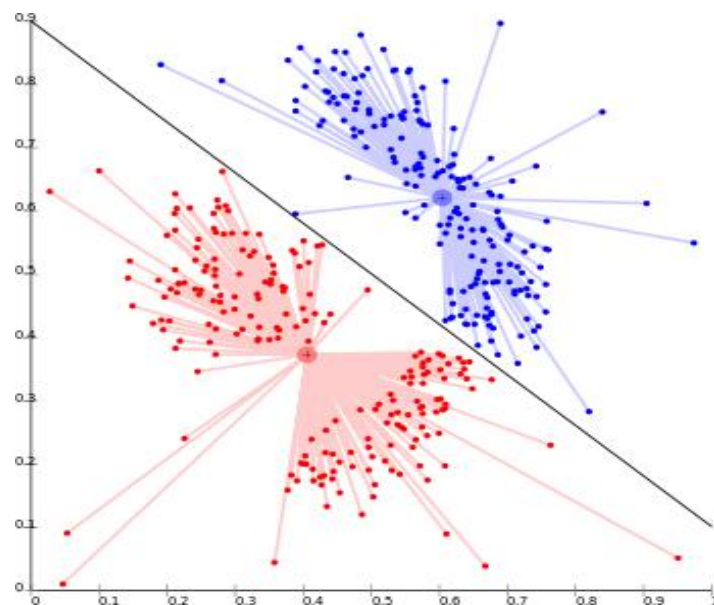


Figure 9: Cluster centers and Cluster distribution of "Wine" dataset

From the Figure 9 we can get that the "Wine" dataset was obvious pointed to two kinds by the fast clustering model based on PSO algorithm Cluster centers and Cluster distribution were given in the Figure 8

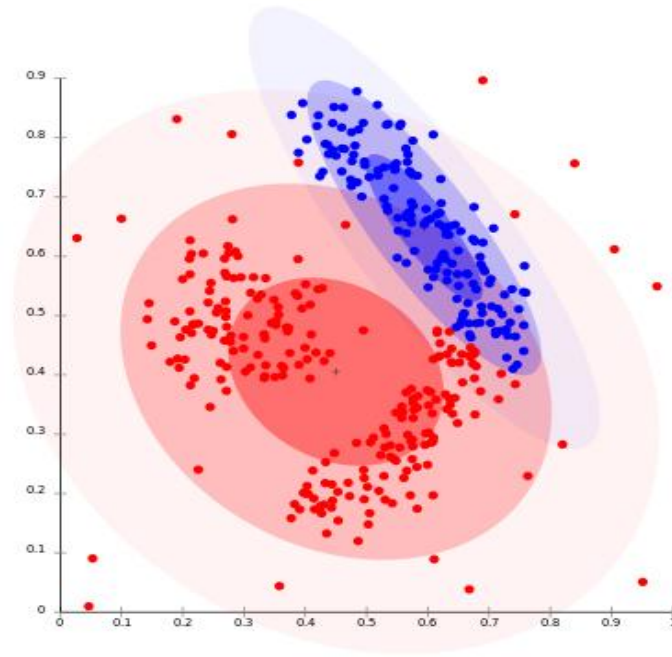


Figure 10: Cluster centers and Cluster distribution of “glass” data

From the Figure 10, The data of “glass” dataset was divided into two groups as the particle swarm. In the graph, the two categories of data are clustered.

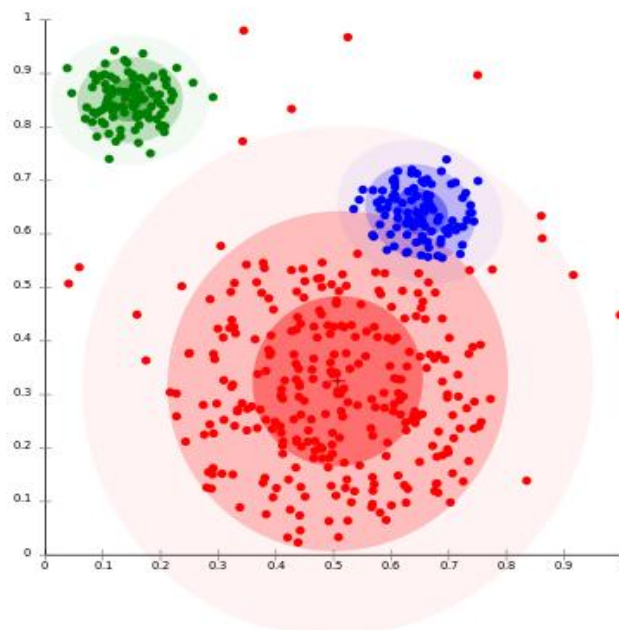


Figure 11: Cluster centers and Cluster distribution of “Aggregation” data

From the Figure 11, we can get that the “Aggregation” data was separated into three groups as the particle swarm. The fast clustering model based on the PSO algorithm accomplished the cluster of “Aggregation” data.

3.2.5 Strength and Weakness

- I Strength:** The clustering model can achieve the goal of categorizing the datasets in an effective, fast and automatic manner. And this clustering model also aims to

make up for the ignorance of the size of datasets which is thought not have much influence on the accuracy of the clustering results. Another giant advantage of the clustering model is that it does not need any prior knowledge. This clustering algorithm belongs to unsupervised learning.

- I **Weakness:** The clustering model needs comparatively good input parameters, such as the threshold value of clustering radius. And also the fitness function is sensitive to two penalty factors.

IV. Conclusions

4.1 Conclusions of the Problems

- I For the PTTRISM, the vector space of lexical meaning and word frequency analysis demonstrate their practicability in evaluating the risk indexes of people who are monitored on-line and real-time. Through Catanfield datasets validation, and in combination with the real situation, only a very few people are likely to become a potential terrorist by given the set threshold value of the relative risk index (0.1). The result of Catanfield dataset shows that only 12 people from 1398 individuals should deserve for much more attention, because their risk indexes are beyond the set threshold value of the relative risk index.
- I For the MDCSSM, the clustering technique of particle swarm optimization can achieve the goal of categorizing the big monitoring data in an effective, fast and automatic manner. In order to validate the practicability of the Function Model 1, the model2 must be tested by some tough-processing datasets. But due to scarcity of the big monitoring datasets, some other equivalent datasets, such as "Iris", "Wine", "Glass" and "Aggregation" are replaced for the big monitoring data. The experimental results do demonstrate a good performance of clustering datasets in an effective, fast and automatic manner.

4.2 Methods Used in our Models

- I The combination of the vector-space of lexical meaning and word-frequency analysis promotes the reliability of web text content to evaluate the relative risk index. The adoption of the entropy weight based on information retrieval enhances the analysis of the internet visitors' typical behaviors.
- I The unsupervised learning technique of density-based clustering is utilized to separate the categories of datasets, in order to accelerate the computing time, the standard particle swarm optimization with the weight-shrink factor is adopted in this paper, and more significantly, based the essential thoughts of particle swarm optimization, the clustering procedures resembles the optimization procedures to achieve the goal of categorizing the informational dataset in an effective, fast and automatic manner.

4.3 Applications of our Models

- I The model 1 does not only help evaluate the internet visitors' behaviors and detect, even discriminate the visitor's identity, but also provide a robust tool for commercial people that can be applied to dig out potential customers.
- I The model 2 is applicable to many other subjects, besides data mining. Some academic researchers have already successfully applied k-means clustering to image processing, such as image segmentation, edge-detection. This clustering

algorithm can replace the k-means clustering in image processing.

V. Proposal to Fighting Terrorism

From the statistics of the annual number of terrorists' attack from 1968 to 2009, the number of terrorists' attack is increasing year by years. The terrorist attacks have caused a bad influence on world peace. Many countries in the world have committed to carry out the activities to fighting terrorist organizations. For example, Russian dispatched fighters to bomb ISIS, a representative of world terrorist organizations. Thus it is imperative to fight terrorists and prevent. There are three strategies that may help fight terrorism as follows:

I Put into more resources in terms of network against terrorism. You could build User Online Monitoring System of Behavior and Psychological to monitor and assess the behavior of the public. It's found that only few people has tendency of terrorism. You could establish network information security evaluation system to weaken or even prevent the terrorist propaganda through the network.

I It is a feasible way to unit global network system to monitor online users' behavior. Because terrorists or people who have a tendency to terrorism spread around the world, particularly in the Middle East and Afghanistan. By this way, we can master the action of terrorism organizations wherever terrorists are. All countries around the world should stand in the same line in the aspect of fighting terrorism.

I Strengthen public anti-terrorism education, raise public awareness of anti-terrorism. On the one hand, it could reduce the number of people that are recruited by terrorist organizations; on the other hand, it might be able to reduce the number of casualties of terrorist attack.

VI. References

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